

# The preimage ideal of the compact operators

`MathlibAnnex.CStarAlgebra.compactPreimageIdeal`

**def**

Defines the two-sided ideal  $J_\rho = \rho^{-1}(\mathcal{K}(H))$ .

## Statement

Let  $A$  be a complex  $C^*$ -algebra, with no unit assumed, let  $H$  be a complex Hilbert space, and let  $\rho : A \rightarrow \mathcal{B}(H)$  be a  $*$ -representation. Define

$$J_\rho = \rho^{-1}(\mathcal{K}(H)) = \{a \in A : \rho(a) \text{ is compact}\}.$$

The declaration packages this set as a two-sided ideal of  $A$ .

## Definition

The ideal laws follow from

$$\rho(0) = 0, \quad \rho(a + b) = \rho(a) + \rho(b), \quad \rho(-a) = -\rho(a), \quad \rho(ab) = \rho(a)\rho(b),$$

together with the facts that compact operators are closed under sums, negatives, and composition on either side by a bounded operator.

## Assumptions

No unit, separability, irreducibility or faithfulness assumption is used.

## Conclusion

The result is the algebraic two-sided ideal  $J_\rho$ . The definition does not assert norm closedness here.

The later theorem Norm closedness of the compact-preimage ideal proves norm closedness:  $\rho$  is contractive and hence continuous, and  $I_\rho = \rho^{-1}(\mathcal{K}(H))$  is the inverse image of the norm-closed set of compact operators. This observation is kept separate from the algebraic construction.

## Main citations

- Exact definition and context
- Norm closedness of the compact-preimage ideal

## Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
-- The inverse image of the compact operators under a star representation,
as an algebraic two-sided ideal. -/
def compactPreimageIdeal [NonUnitalCStarAlgebra A]
  (rho : A →*na[C] (H →L[C] H)) : TwoSidedIdeal A :=
  TwoSidedIdeal.mk' {a | rho a ∈ compactOperator (RingHom.id C) H H}
  (by
    change rho 0 ∈ compactOperator (RingHom.id C) H H
    rw [map_zero]
    exact Submodule.zero_mem _)
  (fun {x y} hx hy => by
    change rho x ∈ compactOperator (RingHom.id C) H H at hx
    change rho y ∈ compactOperator (RingHom.id C) H H at hy
    change rho (x + y) ∈ compactOperator (RingHom.id C) H H
    rw [map_add]
    exact Submodule.add_mem _ hx hy)
  (fun {x} hx => by
    change rho x ∈ compactOperator (RingHom.id C) H H at hx
    change rho (-x) ∈ compactOperator (RingHom.id C) H H
    rw [map_neg]
    exact Submodule.neg_mem _ hx)
  (fun {x y} hy => by
    change rho y ∈ compactOperator (RingHom.id C) H H at hy
    change rho (x * y) ∈ compactOperator (RingHom.id C) H H
    rw [map_mul]
    have hxy : (rho x).comp (rho y) ∈ compactOperator (RingHom.id C) H H := by
      change IsCompactOperator ↑((rho x).comp (rho y))
      exact hy.clm_comp (rho x)
    exact hxy)
  (fun {x y} hx => by
    change rho x ∈ compactOperator (RingHom.id C) H H at hx
    change rho (x * y) ∈ compactOperator (RingHom.id C) H H
    rw [map_mul]
    have hxy : (rho x).comp (rho y) ∈ compactOperator (RingHom.id C) H H := by
      change IsCompactOperator ↑((rho x).comp (rho y))
      exact hx.comp_clm (rho y)
    exact hxy)
```

| In the source                              | Mathematical meaning                                                                                            |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| [NonUnitalCStarAlgebra A]                  | A complex $C^*$ -algebra $A$ considered without assuming a unit. This does not assert that a unit cannot exist. |
| (rho : A →* <sub>na</sub> [C] (H →L[C] H)) | The $*$ -representation $\rho : A \rightarrow \mathcal{B}(H)$ .                                                 |
| : TwoSidedIdeal A                          | The output is a two-sided ideal of $A$ .                                                                        |

| In the source                                                | Mathematical meaning                                                                                                               |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <code>{a   rho a ∈ compactOperator (RingHom.id C) H H</code> | Its underlying set is<br>$J_\rho = \{a \in A : \rho(a) \in \mathcal{K}(H)\}.$                                                      |
| <code>TwoSidedIdeal.mk'</code>                               | The constructor supplies the ideal laws for this set.                                                                              |
| <code>map_zero</code>                                        | Because $\rho(0) = 0$ , the zero element belongs to $J_\rho$ .                                                                     |
| <code>map_add</code>                                         | If $a, b \in J_\rho$ , then $\rho(a + b) = \rho(a) + \rho(b)$ is compact.                                                          |
| <code>map_neg</code>                                         | If $a \in J_\rho$ , then $\rho(-a) = -\rho(a)$ is compact.                                                                         |
| <code>hy.clm_comp (rho x)</code>                             | If $y \in J_\rho$ , then $\rho(xy) = \rho(x)\rho(y)$ is compact: a bounded operator composed on the left with a compact operator.  |
| <code>hx.comp_clm (rho y)</code>                             | If $x \in J_\rho$ , then $\rho(xy) = \rho(x)\rho(y)$ is compact: a compact operator composed on the right with a bounded operator. |

Exact source and proof.

Earlier published Card and PDF

#### ▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.compactPreimageIdeal`

Accepted content SHA-256: `dc3b994b4cd43b9806c2a04378c1261b000f9df6f30e545a54cb93ac7e4cc361`

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Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`